

Ride-Hailing Application (Rapido/Uber/Ola) – User & Driver Registration and Verification Workflow

Document Version

Version: 1.0

Purpose: This document describes the complete registration, identity verification, document verification, and approval workflow for a ride-hailing mobile application.

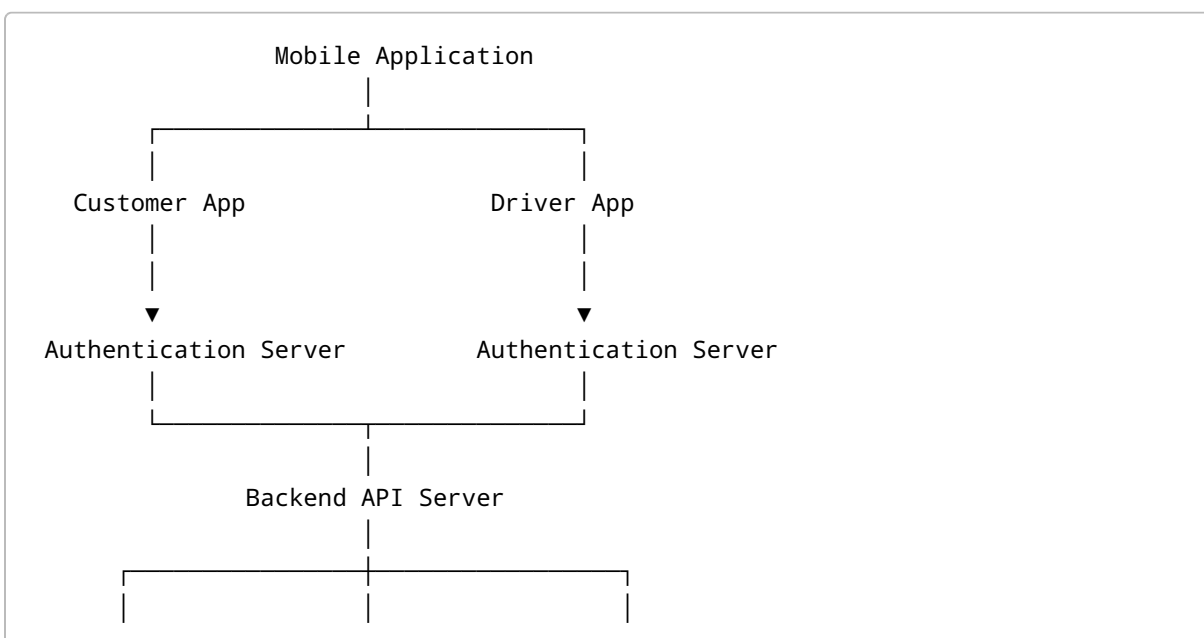
1. Introduction

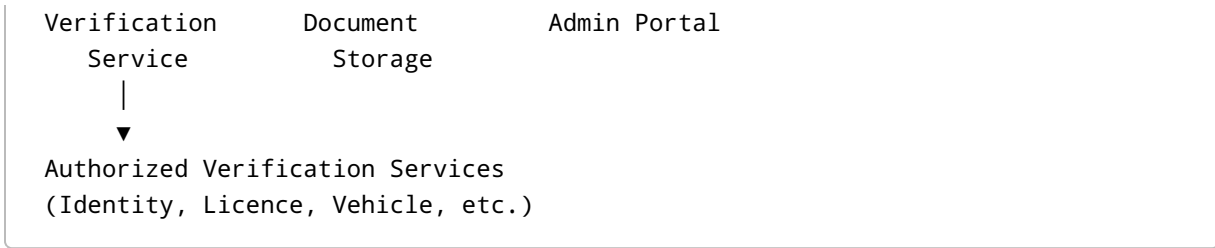
A ride-hailing application consists of three major stakeholders:

- Customer (Passenger)
- Driver (Captain/Rider)
- Administrator

Before a driver is allowed to accept ride requests, the platform must verify both the driver's identity and the vehicle documents. This process ensures passenger safety, legal compliance, and fraud prevention.

2. Overall Registration Architecture





3. Customer Registration Workflow

Step 1 – Mobile Number Verification

Customer enters:

- Mobile Number

↓

OTP is sent

↓

Customer enters OTP

↓

OTP Verified

Step 2 – Profile Creation

Customer enters

- Full Name
- Email (Optional)
- Profile Picture (Optional)

↓

Profile Created

Step 3 – Payment Setup

Customer may add

- UPI
- Debit Card
- Credit Card
- Wallet
- Cash

↓

Registration Complete

4. Driver Registration Workflow

Driver registration requires multiple verification stages.

Stage 1 – Account Creation

Driver enters

- Mobile Number

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OTP Verification

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Create Password

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Basic Profile Created

Stage 2 – Personal Information

Driver enters

- Full Name
- Date of Birth

- Gender
- Address
- Emergency Contact
- Email



Proceed to Identity Verification

5. Identity Verification

Driver uploads

- Aadhaar
- PAN
- Selfie



System stores documents securely



Documents sent for verification

6. Driving Licence Verification

Driver uploads

Driving Licence



OCR extracts

- Licence Number
- Name
- Date of Birth
- Expiry Date



Verification Service checks

- Licence exists
- Licence active
- Licence expired?
- Vehicle category valid



Verification Result

Approved / Rejected

7. Vehicle Registration

Driver enters

Vehicle Details

- Vehicle Number
- Manufacturer
- Model
- Year
- Fuel Type



Upload Vehicle Documents

8. Vehicle Documents Required

Driver uploads

- Registration Certificate (RC)
 - Insurance Certificate
 - Pollution Under Control (PUC)
 - Vehicle Photos
 - Number Plate Photo
-

9. Vehicle Verification Process

OCR extracts

- Registration Number
- Owner Name
- Chassis Number
- Engine Number

↓

Verification Service validates

- Vehicle exists
- RC valid
- Registration active
- Owner matches
- Vehicle type matches

↓

Vehicle Approved

10. Insurance Verification

OCR extracts

- Policy Number
- Vehicle Number
- Expiry Date

↓

Verification Service

Checks

- Insurance active
- Policy valid
- Vehicle number matches

↓

Approved

11. Pollution Certificate Verification

OCR extracts

- Certificate Number
- Vehicle Number
- Expiry Date

↓

Verification

Checks

- Certificate valid
- Not expired

↓

Approved

12. Selfie Verification

Driver captures Live Selfie

↓

AI performs

- Face Detection
- Liveness Detection
- Anti-Spoof Detection
- Face Matching

↓

Face compared with

- Aadhaar Photo
- Driving Licence Photo

↓

Match Score Generated



Approved

13. Vehicle Image Verification

Driver uploads

- Front View
- Rear View
- Left View
- Right View
- Number Plate



AI checks

- Vehicle Model
- Vehicle Color
- Number Plate
- Image Quality
- Tampering Detection



Approved

14. Background Verification

Platform performs

- Identity Verification
- Driving Licence Validation
- Duplicate Account Check
- Fraud Detection
- Blacklist Check
- Police Verification (where applicable)



15. Manual Verification

Some cases require human review.

Admin checks

- Blurred documents
- Name mismatch
- Fake documents
- Expired documents
- Poor selfie quality
- OCR failures

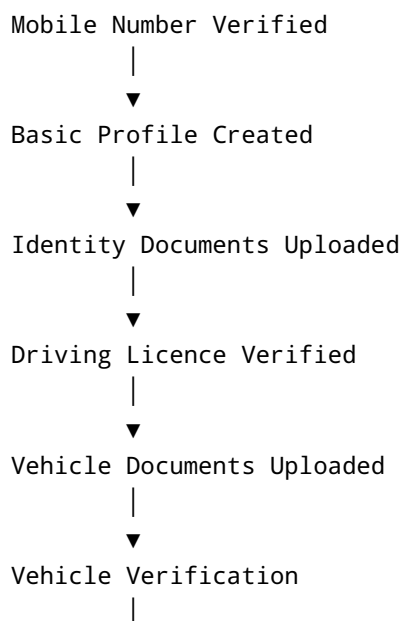
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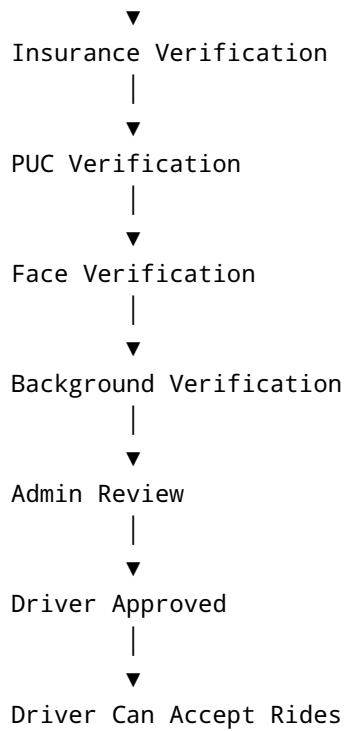
Approve

or

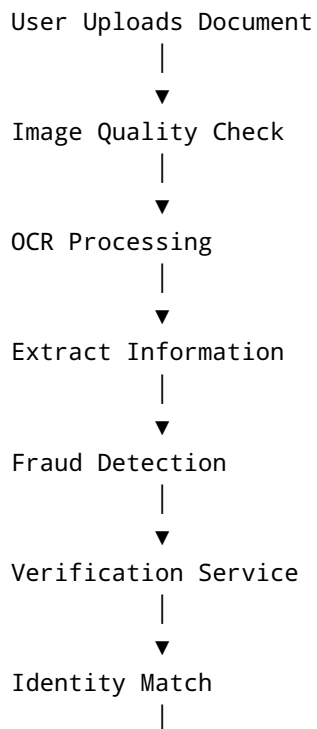
Reject

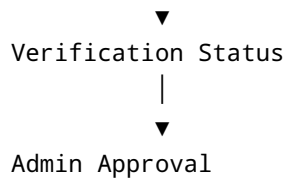
16. Final Approval Workflow





17. Backend Verification Workflow





18. Database Tables

Customer

- CustomerID
- Name
- Mobile
- Email
- ProfilePhoto
- RegistrationDate

Driver

- DriverID
- Name
- Mobile
- Email
- Address
- Status

Aadhaar Verification

- VerificationID
- DriverID
- DocumentID
- VerificationStatus
- VerifiedDate

PAN Verification

- DriverID
 - PAN Number
 - VerificationStatus
-

Driving Licence

- DriverID
 - Licence Number
 - Expiry Date
 - Status
-

Vehicle

- VehicleID
 - DriverID
 - Vehicle Number
 - RC Number
 - Insurance Number
 - PUC Number
 - Status
-

Verification Log

- LogID
 - DriverID
 - Verification Type
 - Result
 - Timestamp
-

19. Admin Dashboard Features

Administrator can

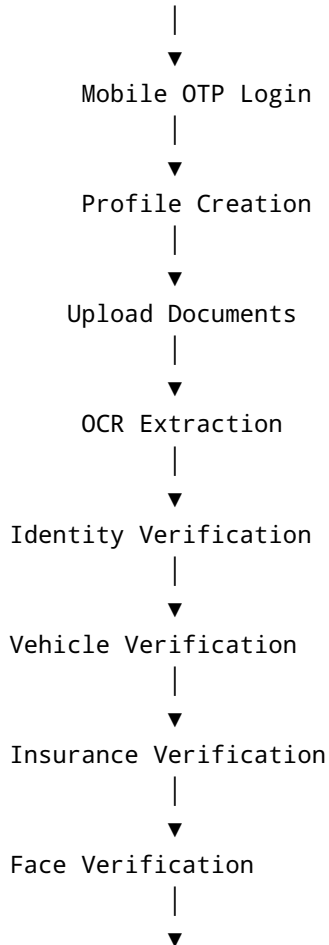
- View pending registrations
 - View uploaded documents
 - Verify driver identity
 - Approve drivers
 - Reject applications
 - Suspend drivers
 - Reactivate drivers
 - View verification logs
 - Search drivers
 - Monitor expired documents
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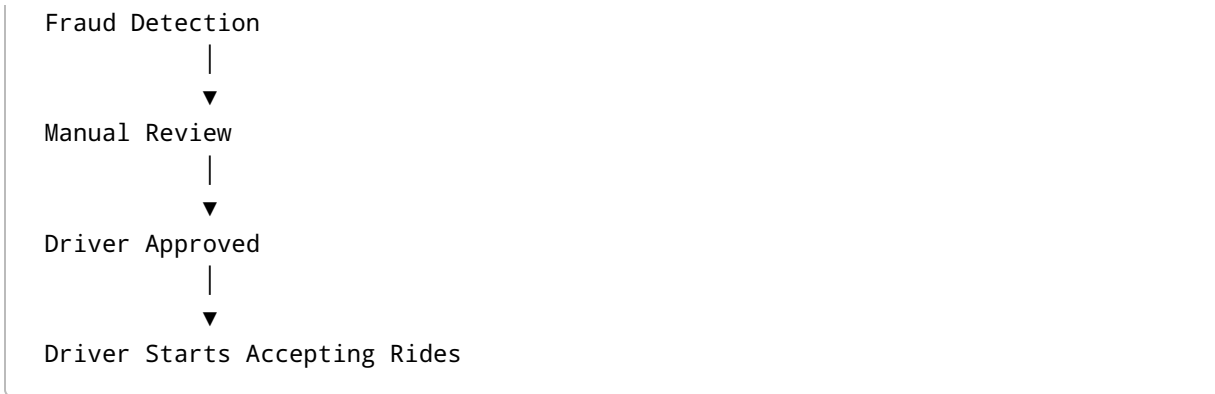
20. Security Measures

- OTP-based authentication
 - Encrypted document storage
 - HTTPS communication
 - Role-based access control
 - Audit logs
 - Secure session management
 - Fraud detection
 - Duplicate account detection
 - Liveness detection
 - Face matching
 - Data encryption at rest and in transit
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21. Complete Registration Summary

Customer / Driver Opens App





22. Conclusion

A modern ride-hailing application follows a layered verification process combining OTP authentication, document upload, OCR-based data extraction, AI-assisted image analysis, identity verification through authorized verification services, manual review for exceptional cases, and final administrative approval. This multi-step approach helps ensure that only verified drivers and compliant vehicles are permitted to operate on the platform while providing a secure experience for passengers.